

EXPERIMENT NO.8

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Batch:B

Aim: To implement the Blockchain platform Ganache and deploy a simple Solidity smart contract using Metamask and Remix IDE.

Theory:

1. Introduction to Blockchain

Blockchain is a **distributed, decentralized digital ledger** that records transactions across multiple nodes in a network. Each transaction is stored in a block, and blocks are linked together using cryptographic hashes, forming a secure and immutable chain.

Key features:

- **Decentralization** – No central authority controls the system
- **Immutability** – Data cannot be altered once recorded
- **Transparency** – Transactions are visible to all participants
- **Security** – Uses cryptographic techniques

2. What is Ganache?

Ganache is a **personal blockchain simulator** used for developing, testing, and debugging Ethereum-based decentralized applications (DApps).

It is part of the **Truffle Suite**, which provides tools for smart contract development.

Ganache creates a **local blockchain environment** that mimics the behavior of the real Ethereum network.

3. Features of Ganache

a. Local Blockchain

Ganache runs a blockchain locally on your system, allowing developers to test without using real cryptocurrency.

b. Pre-funded Accounts

- Provides multiple accounts with preloaded Ether
- Useful for testing transactions without cost

c. Instant Mining

- Transactions are mined instantly
- No waiting time like real networks

d. Detailed Transaction Logs

- Displays:
 - Transaction hash
 - Gas usage
 - Block number
 - Events

e. Smart Contract Testing

Supports deployment and testing of smart contracts written in Solidity.

f. Customizable Settings

- Gas limit
- Gas price
- Number of accounts
- Network ID

4. Architecture of Ganache

Ganache simulates the Ethereum architecture:

a. Accounts Layer

- Contains multiple Ethereum accounts
- Each account has:
 - Address
 - Private key
 - Balance

b. Blockchain Layer

- Stores blocks and transactions
- Each block contains:
 - Block number
 - Timestamp
 - Transactions

c. Virtual Machine

Ganache uses the **Ethereum Virtual Machine** to execute smart contracts.

d. RPC Server

- Provides JSON-RPC API
- Allows interaction using tools like Web3.js

5. Working of Ganache

1. Ganache starts a local blockchain network
2. It generates multiple accounts with Ether
3. Developers deploy smart contracts
4. Transactions are executed
5. Ganache mines blocks instantly
6. Results are displayed in logs

6. Advantages of Using Ganache

- No real Ether required
- Faster testing (instant mining)
- Easy debugging with detailed logs
- Safe environment (no risk of real loss)
- Ideal for beginners and developers

7. Limitations of Ganache

- Not a real decentralized network
- Does not simulate real-world latency
- Limited scalability testing
- Not suitable for production deployment

8. Applications of Ganache

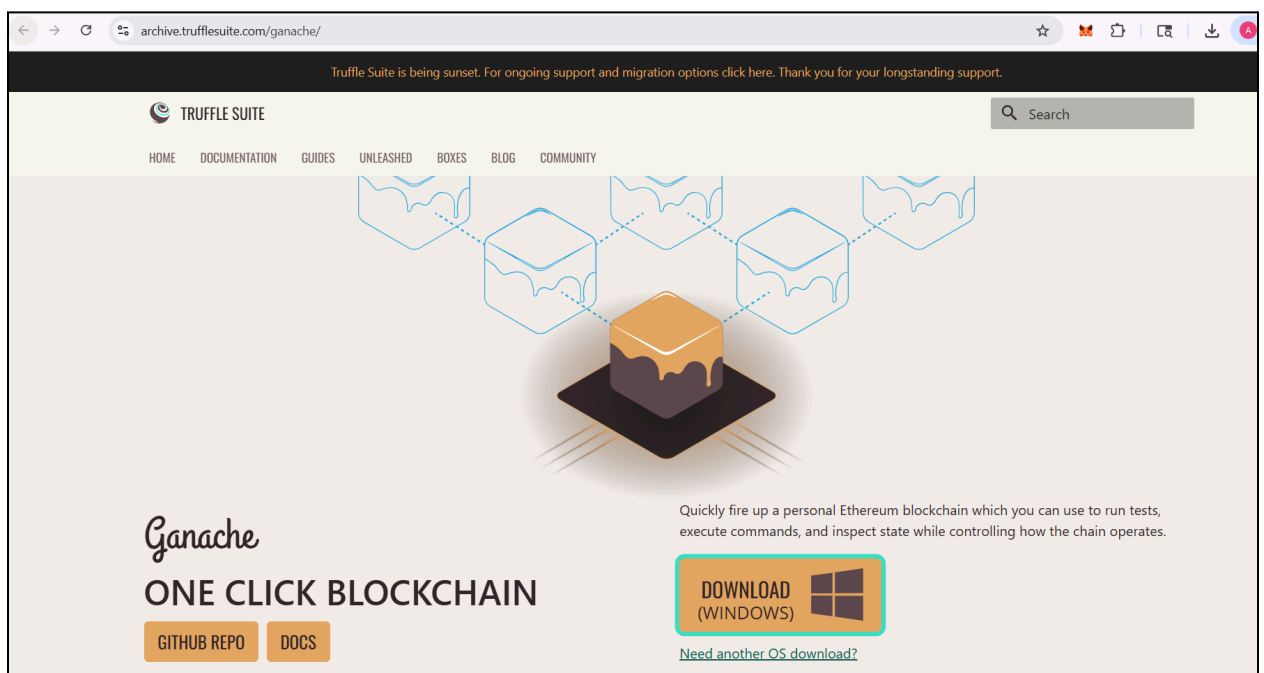
- Smart contract development
- Testing decentralized applications (DApps)
- Learning Ethereum blockchain concepts

- Debugging transactions

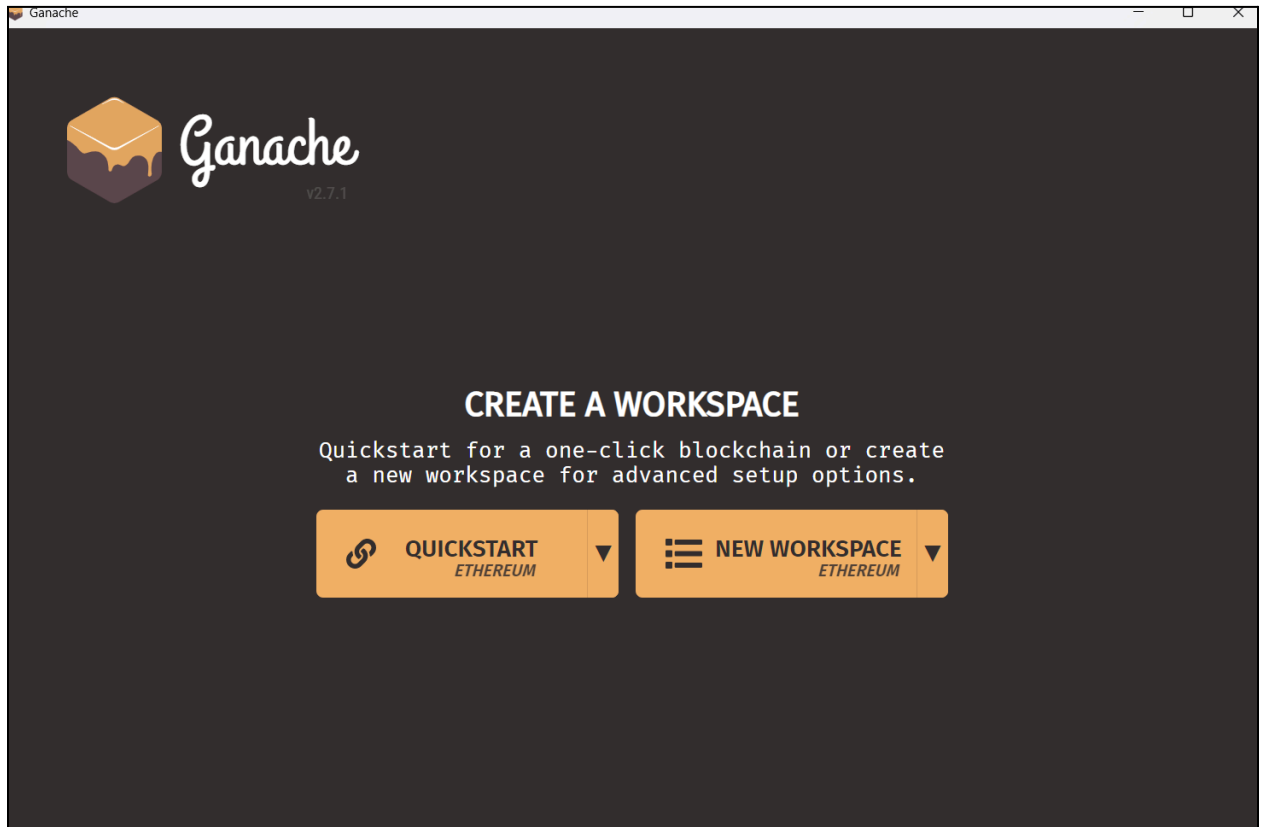
Implementation:-

1. Set Up Ganache (The Local Blockchain)

1. **Download & Install:** Go to the [Truffle Suite website](https://trufflesuite.com/ganache/) and install Ganache.



2. **Launch:** Open Ganache and select "Quickstart".



3. **Identify Details:** You will see a list of 10 addresses, each with 100 ETH.

- Look at the top bar for **RPC Server**. It usually says **HTTP://127.0.0.1:7545**.
- Look at the **Network ID**. It is usually **5777**.

ACCOUNTS

BLOCKS

TRANSACTIONS

CONTRACTS

EVENTS

LOGS

SEARCH FOR BLOCK NUMBERS OR TX HASHES

CURRENT BLOCK0

GAS PRICE20000000000

GAS LIMIT6721975

HARDFORKMERGE

NETWORK ID5777

RPC SERVERHTTP://127.0.0.1:7545

MINING STATUSAUTOMINING

WORKSPACE QUICKSTART

SAVE

SWITCH

MNEMONIC

milk enforce ritual climb juice dog super velvet alone spray banner season

HD PATH

m44'60'0'0account_index

ADDRESS

0x42Ddeef357500e75EE717cD0d2e54adF43D4bB7F

BALANCE

100.00 ETH

TX COUNT

0

INDEX

0

ADDRESS

0x49D75d56Eb9F227640910be560d0c254060e852e

BALANCE

100.00 ETH

TX COUNT

0

INDEX

1

ADDRESS

0xAed3a4Fc5c857C5F094DA4A40e493638910DA3eF

BALANCE

100.00 ETH

TX COUNT

0

INDEX

2

ADDRESS

0x964a05d1fD58941d894129d4e28CAcf7D7F89c88

BALANCE

100.00 ETH

TX COUNT

0

INDEX

3

ADDRESS

0xa4642F190Ae5Ef8c0Edc7E87395655063cf30b1E

BALANCE

100.00 ETH

TX COUNT

0

INDEX

4

ADDRESS

0xE0495F9ECECB322c76D289a7A33d58645f0b6b9B

BALANCE

100.00 ETH

TX COUNT

0

INDEX

5

2. Connect Ganache to MetaMask

MetaMask doesn't know about your local Ganache "server" yet, so we have to add it manually.

1. Open **MetaMask** -> Click the **Network Dropdown** (top left) -> **Add Network**.
2. Select **"Add a network manually"** at the bottom.
3. Enter these details:
 - **Network Name:** Localhost 8545
 - **New RPC URL:** **http://127.0.0.1:7545**
 - **Chain ID:** **1377**

- **Currency Symbol:** ETH

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Add a custom network

×

Network name

Localhost 8545

Default RPC URL

Ganache RPC
127.0.0.1:7545

Chain ID

1337

Currency symbol

ETH

4. **Import Account:**

- In Ganache, click the **Key Icon** on the far right of the first address.

ADDRESS	BALANCE	TX COUNT	INDEX	
0x7110a5Ed96696D2FAA63C5F84f69Fc71B984DBfd	100.00 ETH	0	0	

- Copy the **Private Key**.

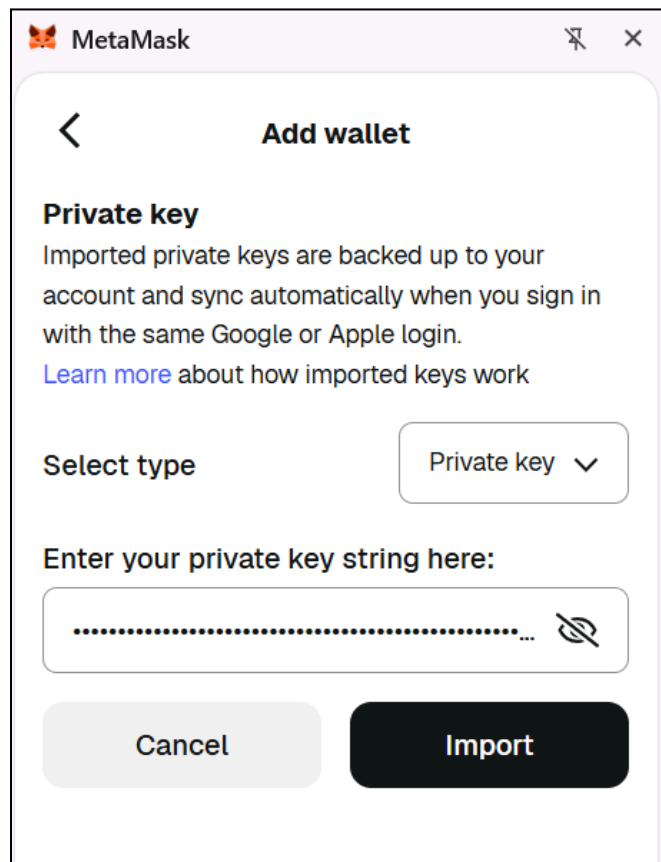
ACCOUNT ADDRESS

0x7110a5Ed96696D2FAA63C5F84f69Fc71B984DBfd

PRIVATE KEY

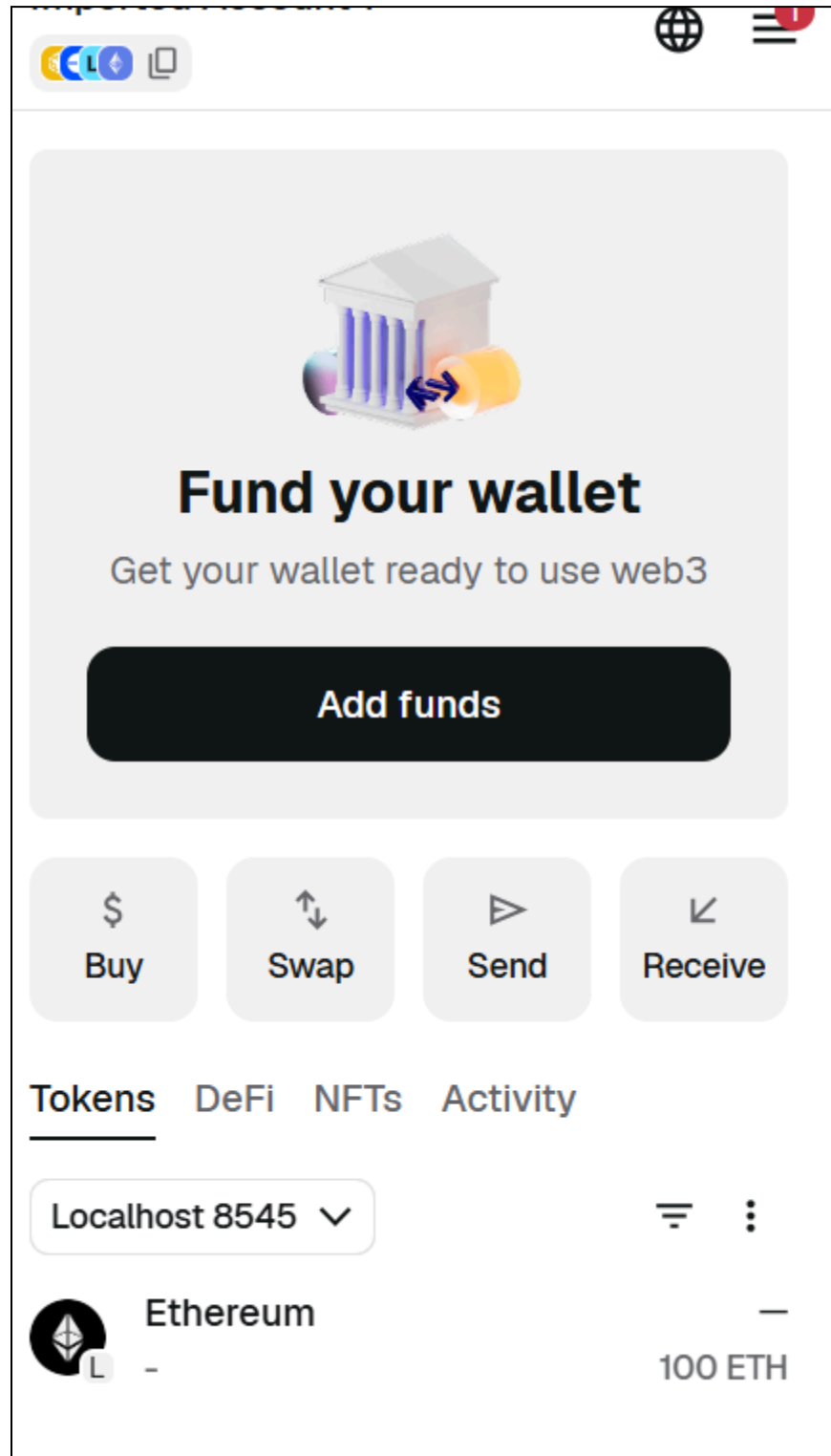
0xf4a7a11cb7e2aa839bae7d439672da8684ee9d061f2b809eb0c0da4700e845

- In MetaMask, click the Account Circle -> **Import Account** -> Paste the Private Key.



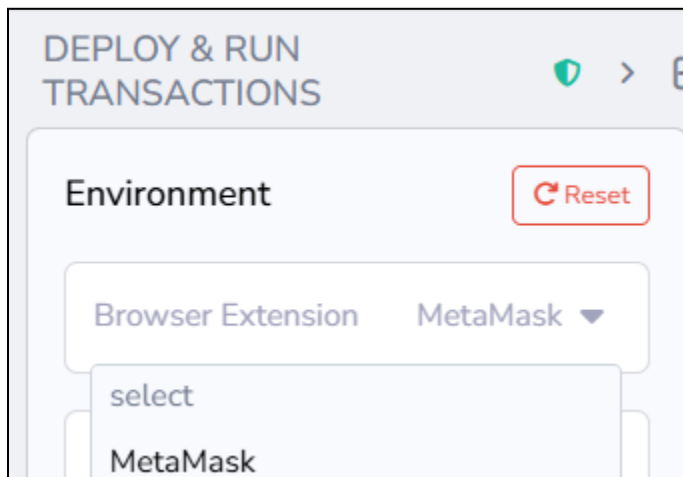
The screenshot shows the MetaMask application window with the title bar 'MetaMask'. Inside, the 'Add wallet' screen is displayed. At the top left is a back arrow, and at the top center is the title 'Add wallet'. Below this, the section 'Private key' is highlighted. The text explains that imported private keys are backed up to the account and sync automatically. A link 'Learn more about how imported keys work' is provided. Under 'Select type', a dropdown menu shows 'Private key' with a downward arrow. Below this, the instruction 'Enter your private key string here:' is followed by a text input field containing a series of dots and a toggle icon (an eye with a slash). At the bottom, there are two buttons: 'Cancel' (light gray) and 'Import' (dark gray).

- You should now see **100 ETH** in MetaMask!

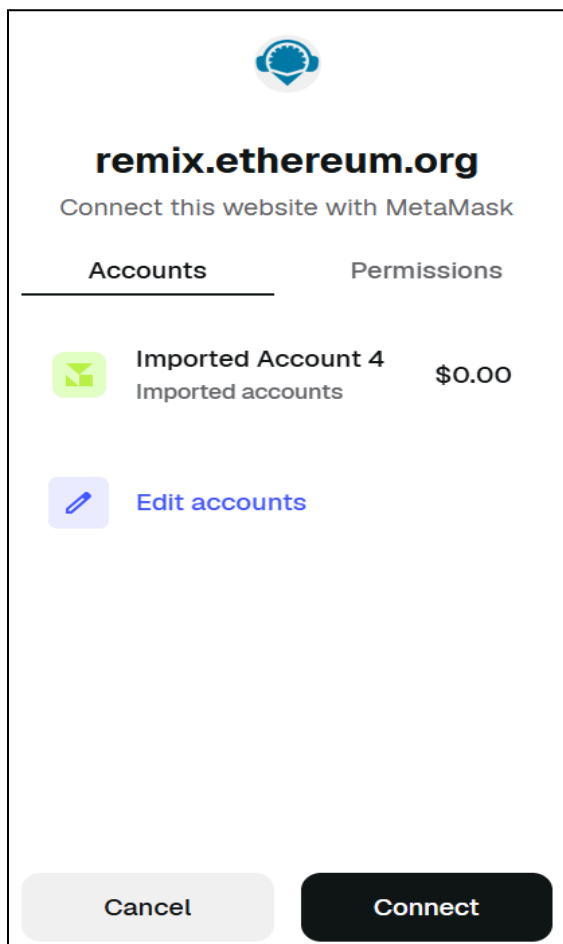


3. Connect Remix to MetaMask (Injected Provider)

1. In Remix, go to the **Deploy & Run Transactions** tab.
2. Set **Environment** to **Injected Provider - MetaMask**.

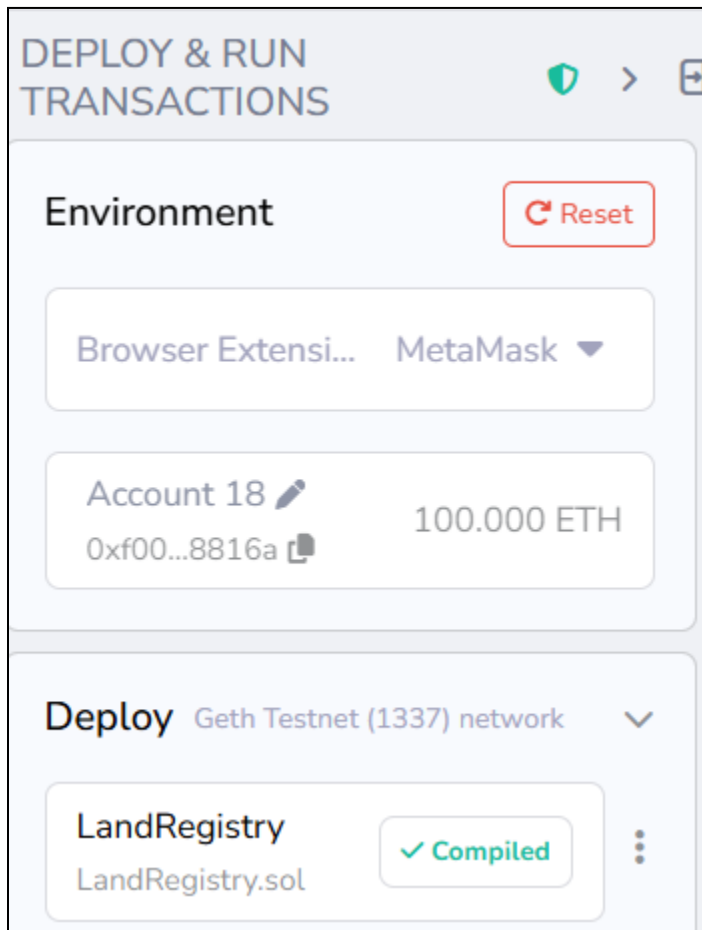


3. MetaMask will ask to connect to "Ganache Local." Click **Confirm**.



4. Compile and Deploy

1. Compile your `LandRegistry.sol`.
2. In the Deploy tab, click **Deploy**.



Deploy Geth Testnet (1337) network

LandRegistry

LandRegistry.sol

✓ Compiled



Verify Contract on Explorers



Value

0

wei



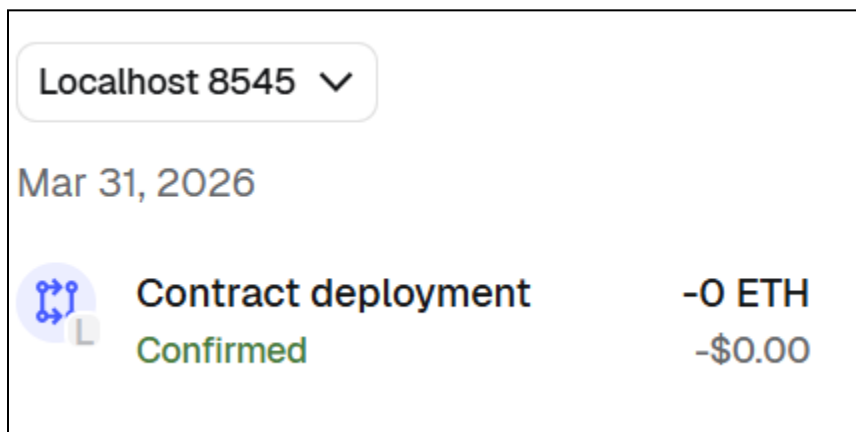
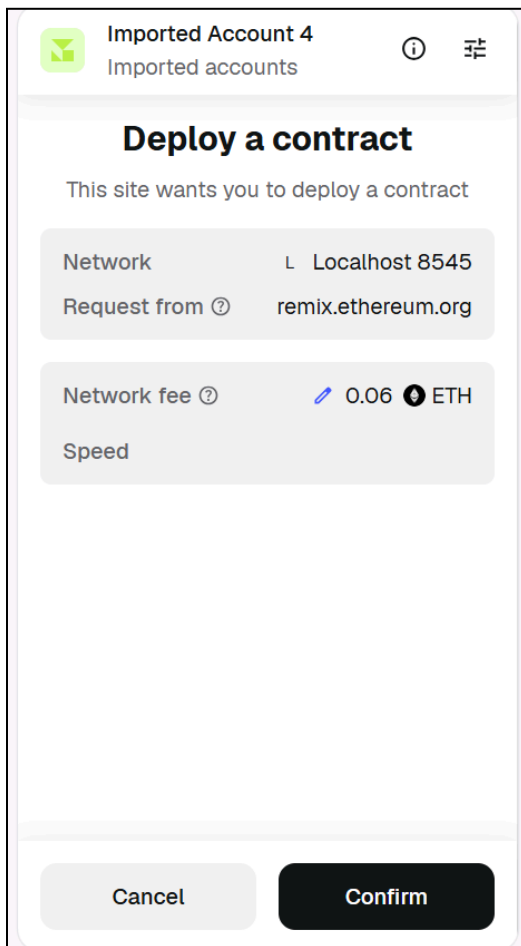
Gas limit

custom

3000000

Deploy

3. MetaMask will pop up. Since this is your local Ganache, the transaction will be **instant** and the gas cost will be deducted from your 100 ETH.



5. Check Transaction Details in Ganache

1. Open the **Ganache Desktop App**.
2. Click the **Transactions** tab at the top.
3. You will see a "Contract Creation" transaction. Click it to see the gas used, the "TX Hash," and the block number

ACCOUNTS

BLOCKS

TRANSACTIONS

CONTRACTS

EVENTS

LOGS

SEARCH FOR BLOCK NUMBERS OR TX HASHES

CURRENT BLOCK

GAS PRICE

GAS LIMIT

HARDFORK

NETWORK ID

RPC SERVER

MINING STATUS

WORKSPACE QUICKSTART

SAVE

SWITCH

TX HASH

0x9b86fee68cee899a31b45d0fb5afdc9a4a8b0e17573bad285a1593d10b7b08bf

CONTRACT CREATION

FROM ADDRESS

CREATED CONTRACT ADDRESS

GAS USED

VALUE

0x7110a5Ed96696D2FAA63C5F84f69Fc71B984DBfd

0x104D6533C9B81cEB346b6808dD28bD3F1dD31547

853999

0

[illegible]

In the **Accounts** tab, we can see that the some balance has been deducted from the first address as part of the transaction that has taken place

CURRENT BLOCK 1	GAS PRICE 20000000000	GAS LIMIT 6721975	HARDFORK MERGE	NETWORK ID 5777	RPC SERVER HTTP://127.0.0.1:7545	MINING STATUS AUTOMINING	WORKSPACE QUICKSTART	SAVE	SWITCH	
MNEMONIC tuna lock spell chat motor sick begin young airport lecture reveal already							HD PATH m44'60'0'0account_index			
ADDRESS		BALANCE		TX COUNT		INDEX				
0x7110a5Ed96696D2FAA63C5F84f69Fc71B984DBfd		99.98 ETH		1		0				
ADDRESS		BALANCE		TX COUNT		INDEX				
0xb9500bADFF07D7D6905122b47DB4511cd3D09208		100.00 ETH		0		1				
ADDRESS		BALANCE		TX COUNT		INDEX				
0x108BeF877BfB9E77B6F3613A7eEdc16030C778dE		100.00 ETH		0		2				
ADDRESS		BALANCE		TX COUNT		INDEX				
0xfD9Af7AA1e63d3554e8E65bB1Dd2f5276Dd19514		100.00 ETH		0		3				
ADDRESS		BALANCE		TX COUNT		INDEX				
0x5a6CC05e5aba194F2969C8E5bAbc66396b0E8E3d		100.00 ETH		0		4				
ADDRESS		BALANCE		TX COUNT		INDEX				
0x16cd8cD29Ec81278B6cf7A73df528EA5108047Ca		100.00 ETH		0		5				
ADDRESS		BALANCE		TX COUNT		INDEX				
0x7868a3579F7e70bba9BFfc2aEA65F857fa94DCD0		100.00 ETH		0		6				

Conclusion:-

The experiment successfully demonstrated the deployment of a **LandRegistry** smart contract on a local blockchain using Ganache, MetaMask, and Remix IDE. By configuring a custom RPC network and aligning the Chain ID, a secure bridge was established to authorize a pre-funded account for transaction gas. This process verified the contract's compilation and its successful migration to a private ledger, confirming the integrity of the decentralized data storage. The setup provides a cost-effective, local environment for testing full-scale blockchain application logic.